Kyle Mackenzie

1mackenziekyle@gmail.com | 1mackenziekyle.github.io | github.com/1mackenziekyle | linkedin.com/in/kdmackenzie

Technical Skills

Software: FreeRTOS, STM32 firmware development, operating systems development, Python & data analysis
Modelling & Controls: Laplace-domain physical system modelling, Simulink, vehicle dynamics, BLDC motor controls
Tools/Environments: CAN, MODBUS, RS-485, Git, Docker, Linux (Debian and Arch), Bash scripting

Relevant Experience

Battery Management Systems Firmware Engineer Co-op May 2025 – Aug. 2025

Moment Energy

- Redesigned and reworked fault-logging architecture in BMS firmware: Reconfigured non-volatile storage of system faults, fault severity levels, and CAN message handling for multi-frame fault strings. Designed, developed, tested, and merged.
- Implemented and debugged the Sunspec Modbus API on an ESP32-based battery pack controller, utilizing the RS-485 protocol.
- Led the research and proof-of-concept of an integration of the clang-format C-style formatting tool to the Texas Instruments IDE, standardizing code style and removing the need for code-style comments on pull requests, thereby speeding up the development cycle.
- Developed and shared a testing tool to emulate the CAN messages sent by a component of the product under different fault scenarios, creating a comprehensive test suite which greatly reduced the testing workflow time for code changes.

Vehicle Dynamics & Controls Developer & Team Lead Sept. 2022 – Dec. 2024

UBC Formula Electric FSAE Design Team

- Led the research and design of four-motor torque vectoring drive algorithm comprised of a sensor fusion algorithm for wheel slip estimation and a cornering response design.
- Built a foundations of a physics-based dynamics model of the car in MATLAB Simulink, to evaluate stability and performance of the controller, and allow year-wide testing, instead of only during Summer months.
- Designed and implemented the first iteration of a software differential for the dual-motor car to improve cornering.

Reliability Test Engineer Co-op May 2024 – Aug. 2024

Corvus Energy

- Developed a certification-level test plan to isolate a single failure mode for power-path connectors.
- Assembled the electrical setup for the busbar components testing including wiring harnessing of busbars and voltage and temperature sensors, and mechanical assembly of the test chamber.
- Analyzed test data to characterize the fatigue failure of busbar components, providing the company with concrete data supporting the lifespan reliability of the product.
- Extrapolated test data and compared with in-field data to get a reliability estimate for deployed battery systems.
- Led meetings while supervisor was away discussing results with senior engineers to make a decision on next steps and key findings.

Battery Controls Systems Co-op Sept. 2023 – Dec. 2023

Corvus Energy

- Tested hardware and software for release of a safety-critical battery management system software version.
- Developed a thermal model of the pre-charge system of the battery pack to control the pre-charge current.
- Streamlined the battery cell data collection process for Lithium-ion cell state-of-health degradation and automated test scripts to validate SOC estimation algorithm.
- Extended test coverage of a hardware-in-the-loop test bench for the battery management system software by implementing the wire harnessing of new sensors and writing new test cases.
- Assisted in the design of a thermistor emulation PCB to extend test coverage of the battery cell simulator test bench.

Technical Projects

OS161 Operating Systems Kernel Development (link) | *C* Sept. 2024 – Dec. 2024

- Implemented high-level synchronization primitives such as locks, condition variables, and semaphores, which were later used to implement thread-safe system calls, a file system, and virtual memory.
- Designed and implemented a thread-safe file system architecture to support file-related syscalls.
- Implemented a virtual memory system that manages a page table to keep track of the state of RAM pages, and a TLB to cache page table entries. Additionally implemented swap-space support.

Capstone Project: Uni-wheeled Robot Drive Controls (link) | *Python* Sept. 2023 – Present

- Won the 2025 Eng Phys Project Fair Eric Roenitz Award for Ingenuity, one of three awards granted to capstone projects.
- Led chassis design to incorporate mechanical design principles, focusing on modularity and rapid (dis)assembly to accelerate the prototype cycle.
- Brought up the software system on a Jetson Nano to facilitate the development of LQR, MPC, and RL controllers.

Machine Learning Self-Driving & Character Recognition (link) | *Python* Jan. 2023 – Apr. 2023

- Developed a novel SAIL (Selective Aggregation for Imitation Learning) method to improve dataset efficiency for imitation learning, selectively adding only informative training samples to enhance model performance.

Education

Bachelor's of Applied Science (Engineering) in Engineering Physics Sept. 2020 - May 2026

University of British Columbia